

Synthesizing Staged Updates between Verified Controllers

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Verified old and new controllers leave the transition unresolved: component states and active requirements must change safely. Fine-Grained Dynamic Update Controller Synthesis (FG-DUCS) takes local transfers, requirement lifetimes and initial obligations as a contract and synthesizes a staged update policy. A winning policy safely completes from every reachable old state under every outcome in finitely many events, preserving the supplied new controller's selected continuation. Commands and handover require states with no enabled uncontrollable event. Contract/game correspondence and established lazy reachability yield a policy with progress rank or a losing-region certificate. A local condition guarantees that separate transfers preserve a merged solution. Other constructed cases require one or the other granularity; merging changes no decision in 54 comparisons derived from existing models. On 27 adapted games, our Modal Transition System Analyser (MTSA) extension decides 26 within 1,200 s using Lazy, versus 17 with Eager, 24 with Update-first and 14 with Direct-Full. Update-first is sometimes faster. Continuous service and elapsed-time bounds are not guaranteed; the intended use is checking finite maintenance plans before deployment.

CCS Concepts: • **Software and its engineering** → **Software evolution**.

Additional Key Words and Phrases: dynamic software updating, discrete-event controller synthesis, supervisory control, on-the-fly synthesis

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1 Introduction

An update must coordinate changing component states and requirements between verified controllers [3, 7, 15, 38]. It should also retain the replacement's allowed continuations, which satisfying new safety goals alone can discard (Section 3).

A workcell may need to move a workpiece before replacing its station. Audit coverage may need overlap; incompatible access rules forbid it. The policy must retain duties across every old state and environment outcome. A fixed sequence can fail under branching; safety alone permits endless updates.

FG-DUCS compiles local choices into a policy for safe finite handover. The contract fixes permitted transfers, requirement lifetimes and endpoint obligations; synthesis chooses how to complete the update (Table 1). Modelers still supply correct models and transfer implementations; synthesis checks these choices together.

Dynamic Update Controller Synthesis (DUCS) [33, 34] and generalized-reactivity(1), or GR(1), updating [2] already generate update control. FG-DUCS fixes both controllers and generates the intervening game from local transfers, requirement boundaries and monitor starting states (Section 2.3). We contribute this construction and its guarantees, without claiming greater expressiveness or a new reachability algorithm.

During maintenance, update control replaces old control. Interval requirements cover gaps between old-stop and new-start. Work may pause, and commands and handover need global uncontrollable (UC) quiescence: no environment event outside the controller's control is enabled. Perpetual UC activity prevents completion; neither continuous service nor elapsed-time bounds follow. Within this scope, synthesis checks a maintenance plan before execution.

Our contributions are:

C1: Contract and game. We prove local-history correspondence and endpoint preservation, and give an invariant-initialization criterion (Sections 3 and 4.1).

C2: Granularity conditions. Obstructions and a local sufficient condition explain when separate choices are necessary or preserve merged feasibility (Sections 2 and 5).

C3: A policy with checkable evidence. Sound and complete lazy synthesis returns a completing policy and rank or a losing-region certificate (Section 6).

C4: Implementation evidence. Reference-game checks and 27 adapted contracts test the MTSA implementation; constructed comparisons separately test granularity (Section 7).

Our public replication package [4], prepared for this paper, includes the cited Technical Appendix A–P and Supplement S1–S5 as Parts I and II of `supplementary_material.pdf`. Data Availability gives versioned links.

2 Why Updates Need Separate Component and Requirement Choices

Verified endpoints leave the intermediate configurations and changing obligations unresolved. Existing update synthesis generates control from a supplied transition model [2, 34]. Two examples explain why that model needs separate component replacements and requirement boundaries. FG-DUCS constructs their combined game from local inputs; Section 2.3 compares this construction with existing interfaces.

2.1 Cell: Moving Work before Replacing a Station

Our constructed *Cell* example has two stations A, B and exactly one workpiece [23, 34]. A station can be replaced only while empty. The old controller moves the workpiece between stations and forbids inspection. The new controller requires inspection at B before departure. Both endpoints are verified, yet simultaneous replacement is impossible: one station always holds the workpiece.

Coordinating moves and changing duties. Figure 1 shows the policy from both old states, called *entries*. Each path replaces the empty station, moves the workpiece, and later replaces the other station. The input requires new-start before old-stop, preventing an unprotected gap; synthesis chooses the remaining order. If B holds at new-start, inspection becomes pending; it must wait until old-stop removes the ban. Starting its monitor—the automaton tracking the new rule—clear would silently discard this duty.

The first path therefore inspects before handover; the second activates the new rule with B empty and needs no inspection. Both reach an allowed new closed-loop handover state with inspection clear: z_A or z_B , according to the holder. A nonnegative *rank* decreases on every arrow, excluding safe but endless inspection. Handover restores the selected endpoint’s full permitted behavior.

What synthesis chooses. These inputs supply a partial constraint, not a complete update sequence. All *Cell* events are controllable: m_A, m_B move to A, B , j inspects, and ρ_A, ρ_B replace the respective empty station. Technical Appendix B gives the complete automata, initializers and path proof. Existing DUCS can encode this staging in its update environment; FG-DUCS generates the staged game from local transfers. The next example shows why requirements also need separate boundaries.

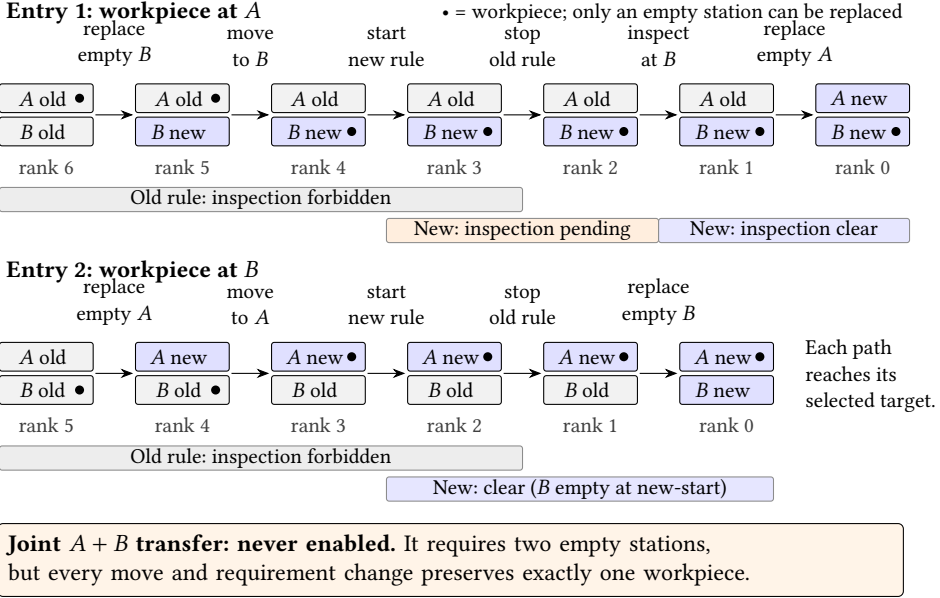
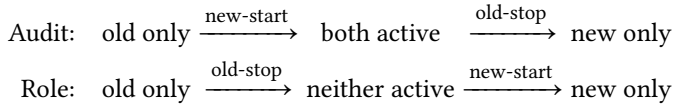


Fig. 1. A complete local policy from both old entries. Bands mark active rules and new-rule inspection memory; new-start precedes old-stop. Each arrow has one action and one outcome, decreasing rank toward its selected new target. The failed joint replacement uses the same endpoints and requirements.

2.2 Audit and Role Lifetimes

We call the second constructed example *Policy(2)*: the number denotes its two old/new requirement pairs, audit and access role. At least one audit rule must remain active throughout, while the old and new roles must never overlap. These obligations require opposing boundary orders:



The role gap is declared by the interval requirement, rather than forced by access rules alone. The new audit monitor starts unlogged and requires a log before service. Each endpoint alternates its own log and service, permitting only its own access role. With an identity transfer, the saved policy obeys these opposing orders and reaches a compatible new endpoint from both old states in at most six events.

If the starts are merged and the stops are merged, the first start overlaps roles and the first stop removes audit coverage. Ordinary events and transfer do not change which rules are active, so neither old entry has a completing merged policy. Technical Appendix C gives the complete finite input, policy and losing closure. This compares merged and separate FG-DUCS requirement boundaries.

2.3 Existing Update Interfaces

Earlier endpoint reconnection. On-the-fly DUCS (O-DUCS) separates update and post-update analysis [18, Sec. 3]. The archived fork in S4.8 synthesizes a new controller, matches component and new-safety states, and copies its transitions [4]. Identity with the 2024 release is unestablished.

Table 1. Supplied inputs and generated results: DUCS [34, Defs. 4.1–4.3, Sec. 5], GR(1) updating [2, Def. 3.1, Sec. 4], and FG-DUCS.

Interface	Supplied inputs	Generated structure and successful result	Progress and continuation
DUCS	Old controller; old/new environments and goals; global transfer relation; transition requirements	Update environment and goal; controller and entry map covering every old-controller state	After the update trigger: eventually stop, start and reconfigure; enforce new goals from new-start.
GR(1) updating	Old/new GR(1) specifications and switching condition	New controller, bridge and switching bookkeeping	The generated new controller runs unchanged after the bridge.
FG-DUCS	Fixed components/controllers; component transfers; monitors, initializers and precedence; allowed handover states	Mixed-version and boundary game; policy, rank and handover map for every old entry and permitted outcome	Supplied new closed loop is preserved from the selected compatible target.

Independent requirement boundaries. DUCS’s transition requirement T constrains one global stop/start pair, including overlap, gaps and their duties [34, Def. 4.1]. FG separates individual stops and starts/initializers.

Cell through native interfaces. DUCS permits staged reconfiguration inside its supplied new environment E' [34, Sec. 4.2]. Technical Appendices H–I give chosen Cell encodings: DUCS adds eventual handover beyond its safety- T implementation [34, Sec. 5.3]; GR(1) may restrict endpoint edges when synthesizing its new controller. These controllable, deterministic witnesses establish neither general translation nor cost advantage (S5).

3 Defining an Update Contract

3.1 Supplied Choices

Finite LTSs and safety monitors. A labeled transition system (LTS) $\mathcal{L} = (S, \Sigma_{\mathcal{L}}, \delta, s^0)$ has nondeterministic transitions $\delta : S \times \Sigma_{\mathcal{L}} \rightarrow 2^S$. Partition events as $\Sigma = \Sigma_{uc} \uplus \Sigma_c$: controllers may disable controllables but cannot select outcomes or suppress uncontrollable (UC) events. A synchronous-product event requires a transition in every component whose alphabet contains it; other components retain their state [25]. A monitor is an automaton that checks a requirement as events occur. Requirement r has a total deterministic monitor $t_r = (M_r, \Sigma, \eta_r, m_r^0, F_r)$ with absorbing errors for violating prefixes [1, 22]. The empty monitor product has one non-error state. Monitors observe labels; policies also observe successor component states.

Components and transfers. For each $i \in I$, let $e_i^{\text{old}}, e_i^{\text{new}}$ be the pre-update and post-update components and let $g_i \subseteq S_{e_i^{\text{old}}} \times S_{e_i^{\text{new}}}$ be their state-transfer relation. Define $g_i(s) = \{s' \mid (s, s') \in g_i\}$. The supplied g_i includes every modeled transformation outcome. Let $E^v = \parallel_{i \in I} e_i^v$.

Requirements and activation. Let pairwise-disjoint $R^{\text{old}}, R^{\text{new}}, R^{\text{upd}}$ contain the pre-update, post-update, and update-interval safety requirements, respectively. For each $r \in R^{\text{new}} \cup R^{\text{upd}}$, the input supplies a partial *monitor-initialization map* ι_r from version-tagged component tuples to $M_r \setminus F_r$.

Its domain specifies where activation is allowed. Section 4.1 defines history-based and activation-scoped interpretations. For $r \in R^{\text{upd}}$, $\text{dom } t_r$ includes every reachable old component state.

Endpoints and handover targets. For $v \in \{\text{old}, \text{new}\}$, the supplied finite controller C^v preserves uncontrollables and makes $E^v \parallel C^v$ deadlock-free and satisfy R^v . Hiding controller auxiliary events leaves observation alphabet $\bigcup_i \Sigma_{e_i}^v$; the union over both versions is the ordinary alphabet Σ_N . Write Reach for initial-state reachability and

$$\text{CL}^v = E^v \parallel C^v \parallel \parallel_{r \in R^v} t_r.$$

Monitors are passive: every endpoint transition originates in $E^v \parallel C^v$; a monitor only updates on that label and adds no enabled event. The input supplies nonempty, checked targets $Z_{\text{load}} \subseteq \text{Reach}(\text{CL}^{\text{new}})$ with component, controller and new-monitor states.

Update commands and order. The set of update events, each executed exactly once, is

$$\Sigma_{\text{upd}} = \{\rho_i \mid i \in I\} \cup \{\text{stop}(r) \mid r \in R^{\text{old}}\} \cup \{\text{start}(r) \mid r \in R^{\text{new}}\}. \quad (1)$$

These controllable events transfer components, stop old monitors or start new ones, respectively, and are disjoint from ordinary events Σ_N : $\Sigma = \Sigma_N \uplus \Sigma_{\text{upd}}$ and $\Sigma_{\text{upd}} \subseteq \Sigma_C$. Each transfer event ρ_i is one atomic model transition: the controller chooses the update, but cannot choose its g_i outcome.

An optional strict partial order $< \subseteq \Sigma_{\text{upd}} \times \Sigma_{\text{upd}}$ expresses precedence among update operations. For a pending-event set P , event a is order-enabled exactly when $a \in P$ and every $a' < a$ has left P . Monitors self-loop on update events that they do not specify.

Eligible contracts. An eligible contract satisfies the finiteness, typing, totality and non-error conditions above; dependencies form a strict partial order, and endpoint controller alphabets cover environment labels. Eligibility is structural; Section 4.1 justifies initializer meanings.

3.2 States, Transitions, and Completion

States and all entry states. An update state $q = (\xi, \mu, P)$ records each component's version/state $\xi(i) = (v, s)$ with $s \in S_{e_i}^v$, each monitor's state $\mu(t_r) \in M_r \cup \{\perp\}$ ($\perp$ means inactive), and pending commands $P \subseteq \Sigma_{\text{upd}}$. Here $\rho_i \in P$ exactly when component i is old. Old monitors remain active until their stop leaves P ; new monitors become active when their start leaves P ; update monitors are always active.

Admission $\text{Init}(z)$ copies z 's component and old-monitor states, installs interval initializers, leaves new monitors inactive and sets $P = \Sigma_{\text{upd}}$. The update policy replaces C^{old} and its scheduling restrictions. Declared old monitors remain until their stops; undeclared objectives, including old liveness, are not preserved. Duties needed throughout belong in interval monitors. The entry set is

$$Q_0 = \{\text{Init}(z) \mid z \in \text{Reach}(\text{CL}^{\text{old}})\}. \quad (2)$$

All-entry coverage forbids waiting for a favorable old state.

Four kinds of induced transitions. For $\xi(i) = (v, s)$, component i uses e_i^v . The active-component product determines enabled ordinary uncontrollables $U(q)$, including those with unsafe outcomes, independently of updates and goals. For every $a \in \Sigma_{\text{upd}}$, impose

$$U(q) \neq \emptyset \implies \text{Post}(q, a) = \emptyset. \quad (3)$$

This global *uncontrollable quiescence* differs from transaction quiescence [21] and tranquility [42]. Ordinary controls may coexist with UC. The following rules define Post ; unmentioned coordinates stay unchanged. “Advance” means $\mu'(t_r) = \eta_r(\mu(t_r), a)$ for the specified active monitors.

Ordinary event $a \in \Sigma_N$. If a belongs to the local event set of at least one active component and every active component having a can transition, take every synchronous-product outcome and advance every active monitor.

Component update ρ_i . If $U(q) = \emptyset$, order-enabled, and $\xi(i) = (\text{old}, s)$, set $\xi'(i) = (\text{new}, s')$ for each $s' \in g_i(s)$, delete ρ_i from P , and advance every active monitor.

Requirement deactivation $\text{stop}(r)$. If $U(q) = \emptyset$ and order-enabled, advance every active monitor except t_r , retain inactive monitors at $\perp$, set $\mu'(t_r) = \perp$, and delete $\text{stop}(r)$ from P ; t_r does not observe it.

Requirement activation $\text{start}(r)$. If $U(q) = \emptyset$, order-enabled, and $\xi \in \text{dom } t_r$, advance every active monitor, retain other inactive monitors at $\perp$, install $\mu'(t_r) = t_r(\xi)$, and delete $\text{start}(r)$ from P ; t_r begins online observation afterward.

If a precondition is false, the successor set is empty; every nondeterministic outcome is included and environment-chosen. Let $\mathcal{Q}$ contain every state reachable from Q_0 through these transitions. *Safety and handover-ready goals.* An update state $q = (\xi, \mu, P)$ is safe, written $\text{Safe}(q)$, exactly when every active monitor is outside its error states. For $P = \emptyset$ and $z \in Z_{\text{load}}$, write $q \sim z$ when component states (without version tags) and all new-monitor states agree. The handover map κ selects such a full endpoint state z . Goals are

$$Q_{\text{goal}} = \{q = (\xi, \mu, \emptyset) \in \mathcal{Q} \mid \text{Safe}(q) \wedge U(q) = \emptyset \wedge \exists z \in Z_{\text{load}} : q \sim z\}. \quad (4)$$

At a goal, interval obligations end and matched new monitors continue. Handover disables ordinary controllables and installs $\kappa(q)$'s controller state, outside Σ and the rank. UC cannot compete; compatible targets may have different continuations.

State-dependent policies and finite completion. A memoryless update policy is a function $\sigma : \mathcal{Q} \setminus Q_{\text{goal}} \rightarrow 2^{\Sigma_c}$ selecting only events with nonempty successor sets.

The possible next states under this policy are

$$\text{Succ}_{\sigma}(q) = \bigcup_{u \in U(q)} \text{Post}(q, u) \cup \bigcup_{a \in \sigma(q)} \text{Post}(q, a). \quad (5)$$

The environment chooses any uncontrollable and any permitted-event outcome. The returned policy retains one controllable at a quiescent state.

Let $\text{Reach}_{\sigma}(Q_0)$ be the policy closure from Q_0 , stopping at goals.

Runs, winning, and witnesses. A run follows the policy successors and stops at a goal. It is *maximal* if infinite or ending at a goal or a state without successors. A single policy σ is winning for an FG-DUCS instance if, from every $q_0 \in Q_0$, (i) every reachable update state is safe, (ii) every reachable non-goal state has a successor, and (iii) every maximal update run reaches Q_{goal} after finitely many events. This *finite completion* counts events and assumes no fairness. Winning policies exclude controllable self-loops during update; a reached UC self-loop makes a state losing. Ordinary actions can progress, as Cell's moves and inspection do. Handover restores the supplied controller's full matched behavior.

A *progress-ranking function* for a winning policy is $\text{rk} : \text{Reach}_{\sigma}(Q_0) \rightarrow \mathbb{N}$, with value 0 at goal states and satisfying $\text{rk}(q') < \text{rk}(q)$ for every non-goal state q and every $q' \in \text{Succ}_{\sigma}(q)$.

The FG-DUCS problem. Decide whether one policy safely completes the eligible game from every entry under every permitted event and outcome. Return $(\sigma, \text{rk}, \kappa)$ covering all Q_0 , or a losing-region certificate.

3.3 Why the Generated State Represents the Contract

A *local update history* $\ell = z(a_1, o_1) \cdots (a_n, o_n)$ records the old entry, events and realized local successors. Components follow their active-version transitions and transfers; monitors fold their declared active intervals. Commands obey precedence, guards and UC quiescence. A source policy selects controllable labels, retains every UC and permitted outcome, and safely completes every maximal history at a matching quiescent target. Technical Appendix A defines these contract operations on histories.

PROPOSITION 3.1 (CONTRACT/GAME CORRESPONDENCE). *For an eligible contract, map each local update history to its current component tuple, active monitor memories and unexecuted commands. Histories with the same image admit the same next events and all the same local outcomes, agree on safety, and have the same handover targets. This map gives a label- and outcome-preserving correspondence between local update histories and paths of the generated game. A history-dependent policy satisfying the source obligations exists exactly when a winning policy exists in that game.*

PROOF. Fold a source history into its current tuple $F(h)$. At entry this is $\text{Init}(z)$. Each ordinary step uses the same participating component edges and full Cartesian product of outcomes; each transfer uses the same quiescence/order tests and g_i , advances active monitors and consumes its command. Stop removes its own monitor before observation; start installs its initializer after observation by existing monitors. Thus every label and outcome has exactly the same successor image, and safety and matching targets agree.

Induction maps source histories to game paths. Conversely, choose an old tuple witnessing the entry and lift each edge by this one-step correspondence, also for infinite paths. A game policy lifts through F ; applying a source policy to consistently lifted game histories gives a history-dependent game policy. Both directions preserve UC events, all selected-label outcomes, unsafe prefixes, deadlocks and infinite avoiding runs. Finite reachability attractor layers yield a state-based winner exactly when any history-dependent winner exists (the independent memoryless-policy lemma in Technical Appendix A), completing the equivalence. $\square$

This proves correspondence for the supplied contract; requirement meaning and Java frontend correctness remain separate obligations.

PROPOSITION 3.2 (PRESERVATION OF THE SELECTED ENDPOINT). *For a returned WIN certificate and reached goal q , the new-phase submodel reachable after handover at q is label-preserving isomorphic to CL^{new} rooted at $\kappa(q)$. Hence their finite and infinite ordinary-event traces coincide.*

PROOF. Handover retains matching components and monitors and installs the selected controller state. Each new-phase state copies an endpoint state with exactly its labeled transitions, without further policy restriction or return to updating. This bijection preserves and reflects edges, hence finite and infinite paths. $\square$

Continuation properties established at the selected state are preserved; target selection can change which continuation applies.

Why matching and reconnection both matter. Cell's prefix $\rho_B, m_B, \text{start}, \text{stop}, \rho_A$ finishes every command with inspection pending. Physical matching alone admits clear target z_B , whose m_A violates that duty; monitor matching excludes the reset. Conversely, disabling controllable loop b in a one-state endpoint with loops a, b and goal *true* preserves legality, deadlock freedom and safety [34, Defs. 3.9–3.10, 4.3], but loses b -continuations. FG-DUCS's fixed-endpoint contract requires preserving those continuations.

Technical Appendix G characterizes policy reuse under changed endpoints, including a saved production-cell (PC2) instance.

4 What a Winning Policy Guarantees

Initial monitor states must represent current invariants or history-dependent duties. Table 2 separates requirement meaning, game guarantees and execution.

Table 2. Guarantee scopes and their premises. Table 4 maps RQ2 checks to these scopes.

Scope	What follows	Additional premise or boundary
Game	Safe finite completion for all entries/outcomes; matched handover	The supplied transition and endpoint rules (Theorem 6.1)
Invariant from start	Truth at activation and while active	Correct predicates/observations and initializer conformance
Safe prior histories	Continuations preserve obligations of the admitted safe histories	Residual inclusion; an empty history scope makes inclusion vacuous
From update entry	Interval monitors reflect obligations from admission onward	Correct boundary state and entry interpretation; S3 gives sufficient conditions
Execution	Handover within the entry’s event rank; selected endpoint continues	A1–A4; no elapsed-time bound or guarantee of platform conformance

4.1 Giving Requirements Their Intended Lifetimes

An initializer assigns the obligation present at activation; selecting any non-error state is insufficient.

An exact activation obligation. Cell’s owed-inspection bit starts at 1 exactly when B holds the workpiece. Event m_B sets it, j clears it, and m_A is forbidden while it is 1; other labels preserve it. Encoding 0 by c and 1 by d makes every monitor transition identical to this rule, including rejection of forbidden departure. Initialization and suffix induction prove exact enforcement. Figure 1’s first path starts with debt and clears it by ordinary inspection; the second starts clear.

Invariants that begin at activation. Activation-scoped interpretation ACT checks a pure invariant $\Box\varphi$ at the actual post-start observer vector and after every later label. It retains the current truth values of event-tracked properties (fluents), rather than a bit recording a prior violation. For deterministic observers with total update $d : V \times \Sigma \rightarrow V$, define

$$\text{ACT}_\varphi(v) = \{w \mid \forall u \preceq w : \varphi(d^*(v, u))\},$$

where prefixes include the empty word. A faithful safety monitor M has non-error language $S_\varphi = \text{ACT}_\varphi(v_0)$ from its initial state. Let $L_M(m)$ denote its continuations from state m .

PROPOSITION 4.1 (SAFE LOOKUP CELLS FOR PURE INVARIANTS). *If a lookup at v selects a non-error state m reached together with v by a safe word in the unrestricted monitor–observer product, or a state with the same continuation language, then $\varphi(v)$ holds and $L_M(m) = \text{ACT}_\varphi(v)$.*

PROOF. Let h reach the pair (m_h, v) . Faithfulness and non-error reachability give $h \in S_\varphi$, hence $\varphi(v)$. For every suffix w ,

$$w \in L_M(m_h) \iff hw \in S_\varphi \iff w \in \text{ACT}_\varphi(v).$$

The second equivalence uses safety of the prefix h and deterministic observers: future valuations depend only on v and w . Language equivalence transfers the equality to m . $\square$

The witness need not be plant-executable or obey one-shot update restrictions. This is residual-language reasoning [13, 37], applied to the generated lookup. At activation, the adapter must choose the actual post-start key and use the same future observation rules; start is consumed once. Undefined entries and fallback initialization are outside the criterion. A true $\varphi(v)$ alone does not supply a lookup entry.

RQ2 distinguishes finite source checks from conformance at actual activations (Technical Appendix E.1).

When prior histories are part of the requirement. For prefix-closed requirement language S_r , declared activation histories $H_r(\xi)$ and monitor continuations $L_r(m)$, residual soundness (RS) means $L_r(l_r(\xi)) \subseteq \bigcap_{h \in H_r(\xi)} \{w : hw \in S_r\}$. New histories include start once; monitors consume later events. Equality is exact; strict inclusion can reject feasible updates, and an empty history set makes inclusion vacuous. Scope completeness and intent are external obligations (Technical Appendix B.1).

The saved checks use safe pre-activation histories consistent with the current observation for new requirements and a reference beginning at entry for interval requirements. A whole-history interpretation also includes prior violations and is considered separately (S3). Cell's pending initializer is exact for occupancy-based debt but conservative for a start-only reference that starts clear. Safe-history inclusion alone establishes neither the occupancy duty nor the activation-scoped interpretation. The distinction exposes which duties the update must preserve.

4.2 Execution and Handover

The execution connection has four distinct premises (Technical Appendix F):

A1: endpoint observations preserve component/monitor states and same-labeled endpoint edges.

A2: update control disables only unselected controllables, preserves every UC event/outcome, reports the successor and eventually observes a next event.

A3: commands start quiescently and finish atomically, finitely and failure-free with an observed modeled successor.

A4: entry observes/copies the actual old state; quiescent handover installs the selected compatible state, using finite failure-free linearizable boundaries [17].

Under A1–A4, every execution maps to the linked model and reaches handover within the entry rank (Theorem F.1). History completeness places every actual activation history in its declared scope; with RS it justifies the residual interpretation. A2 assumes eventual event progress, without fairness among enabled choices. Internal divergence and unobservable outcomes are excluded; rank bounds events, not time.

5 How Granularity Changes Feasibility

Merging removes intermediate states and can prevent or enable completion.

What is merged. Partition transfers, old stops and new starts separately into blocks. A transfer block $B \subseteq I$ simultaneously permits all targets in $\prod_{i \in B} g_i(s_i)$; a boundary block removes or initializes its monitors at the same component tuple. Other active monitors compose member-label effects, checked to commute pairwise on every monitor/residual state. Mapped precedence must remain irreflexive and acyclic. Ordinary behavior, entries, endpoints, relations, initializers and targets stay fixed; quiescence and all member guards apply. These are comparison contracts, not DUCS encodings.

LEMMA 5.1 (MANDATORY-COMMAND OBSTRUCTION). *Let q_0 be an entry and B a nonempty set of commands pending there. Suppose a safe region R contains q_0 , retains all commands in B , and is closed*

under every safe successor whose label is outside B . If each command in B is disabled or has an unsafe outcome at every state in R , no winning policy covers q_0 .

PROOF. Every goal exhausts pending commands, so a winner must execute a first command in B . Safety and closure keep its prefix in R , where that step is disabled or has an environment-chosen unsafe outcome. Avoiding it yields a non-goal deadlock or infinite run, also losing. $\square$

In Cell’s one-workpiece region, joint transfer is disabled. Policy retains its initial active bits until either grouped boundary violates the interval. PC2-Rolling’s enabled joint transfer violates availability (Section 7.2.1). These results concern specified merges, not native DUCS expressiveness.

A capacity boundary. Readiness gates motivate the constructed Rolling contract [32, 39]. It starts with n ready replicas, no spares, and an interval floor of m ready replicas. A transfer boots its replicas; each becomes ready after one UC event, with no waiting loop. Service self-loops are controllable. The all-new, ready target restores capacity; service may pause during the update.

PROPOSITION 5.2 (ROLLING THRESHOLD). *For $n \geq 2$, $1 \leq m < n$ and $1 \leq k \leq n$, fixed consecutive batches of size k (with a smaller final remainder) admit safe finite completion iff $k \leq n - m$.*

PROOF. Some mandatory batch has size k . Its transfer leaves at most $n - k$ ready replicas, so $k > n - m$ violates the interval. Otherwise, transfer one batch only when all replicas are ready, disable service, and retain every UC readiness outcome before the next transfer. Readiness never falls below $n - k \geq m$. With u old and b booting replicas, $2u + b$ strictly decreases on every selected transfer and every UC event, reaching zero exactly at the target. Thus all maximal runs finish without fairness. $\square$

When separate transfers preserve feasibility. Finer commands need not preserve a merged winner. Two one-state components can synchronize UC labels so that both endpoints are quiescent, but either mixed version has a UC loop. Separate identity transfers then lose; a joint transfer skips the mixed state and wins at rank one (complete contract in Technical Appendix O.12).

PROPOSITION 5.3 (SERIAL TRANSFER PRESERVATION). *Consider transfer-only merging with unchanged entries, endpoints, targets, initializers and ordinary behavior, Cartesian transfer outcomes, commuting active-monitor effects and valid mapped precedence. If every transfer preserves its component’s UC synchronization alphabet and enables no previously disabled local UC label, a merged winner implies a fine winner preserving the same selected endpoint continuations.*

PROOF. The local condition keeps partial batches UC-quiescent. Any partial monitor error would persist to a completed Cartesian outcome, contradicting the merged winner. Finite serial expansion retains completion and endpoint choices; Technical Appendix O.13 gives the full argument. $\square$

The constructed Threads family, with workers sharing a bounded queue, satisfies this condition: old/new worker graphs coincide and transfers preserve idle states and the queue (Technical Appendix O.11). Rolling enables readiness and violates it, but its capacity threshold separately gives safe batch sizes.

6 Synthesizing and Checking an Update Policy

The procedure specializes strong-planning and on-the-fly reachability principles [8, 9, 41], retaining every queried event outcome and producing evidence for every old entry.

A *bucket* is $\text{Post}(q, a)$; empty means disabled. Disabling controls at a UC-enabled state removes runs without creating deadlock, preserving winning (UC priority, Lemma A.1). With quiescent goals and no fairness or controller priority, permitting commands while UC is enabled therefore

adds no winning entries. Changing fairness or priority changes the game. The proof establishes memoryless sufficiency.

6.1 Lazy Successor Generation

Algorithm 1 constructs a reachability attractor [6]. V contains discovered states, X those with all UC candidates queried, and W proved winners. *Promote* adds an unproved safe state in X to W when its nonempty UC successor set lies in W or, at quiescence, one complete nonempty control bucket lies in W . It retains that proving set and assigns rank one above its maximum successor rank. Predecessors are then reconsidered. Lazy queries controls incrementally, keeping unfinished cursors in *Retry*. Failure requires exhausted candidates and no further promotion. Extraction retains lower-rank controls and matching endpoint targets.

Algorithm 1: Lazy

Input: Typed update contract (Section 3)

Output: $(\sigma, \text{rk}, \kappa)$ or (no solution, L)

```

1   $V \leftarrow Q_0$ ;  $X, \text{Retry} \leftarrow \emptyset$ ;  $W \leftarrow V \cap Q_{\text{goal}}$ ;
2   $\text{Open} \leftarrow \{q \in Q_0 \mid \text{Safe}(q) \wedge q \notin Q_{\text{goal}}\}$ ; set  $\text{rk}(q) \leftarrow 0$  for  $q \in W$ ;
3  while true do
4      Add queued goals to  $W$  once with rank zero; exhaust Promote, retaining witnesses/ranks, removing winners
        from Retry and propagating predecessors;
5      if  $Q_0 \subseteq W$  then return  $(\sigma, \text{rk}|_{\text{Reach}_\sigma(Q_0)}, \kappa)$  constructed from the goal witnesses and retained lower-rank
        choices;
6      if  $\text{Open} \neq \emptyset$  then
7          Remove  $q$  from  $\text{Open}$ , add it to  $X$ , and query all uncontrollable candidates;
8          Close the control cursor if UC-enabled; otherwise initialize it and query to the next nonempty bucket or
            exhaustion;
9          Insert  $q$  into Retry iff  $q \notin W$  and its cursor has remaining candidates;
10     else
11         if  $\text{Retry} = \emptyset$  then return (no solution,  $L \leftarrow V \setminus W$ );
12         Remove  $q$  from Retry; query to the next nonempty bucket or exhaustion; reinsert iff  $q \notin W$  and its
            cursor has remaining candidates;
13     Add every queried successor to  $V$ ; queue newly discovered goals for rank zero and safe non-goals for  $\text{Open}$ 
        (unsafe states stay outside  $W$  and  $\text{Open}$ );
```

6.2 Correctness and Certificates

THEOREM 6.1 (TERMINATION, SOUNDNESS, AND COMPLETENESS). *For every eligible finite FG-DUCS instance with Equation (3) and quiescent goals in Equation (4), Algorithm 1 terminates. A returned $(\sigma, \text{rk}, \kappa)$ keeps every active monitor non-error, preserves every uncontrollable event and every outcome of each policy event, reaches Q_{goal} finitely without deadlock, and gives each reached goal a compatible target in Z_{load} for the supplied game. It returns no solution iff no winning policy covers Q_0 in that supplied game.*

PROOF. *Coverage.* Every new safe non-goal enters Open and leaves only after expansion queries all UC candidates. At quiescence, finite control cursors retain unfinished nonwinners in *Retry*; promotion removes only winners. Thus expanded states have complete UC outcomes, unfinished nonwinners remain retryable, and unexpanded safe non-goals remain open.

Termination. Each state is expanded/promoted once and each candidate queried once. Reverse incidences are processed on promotion, so the finite search terminates.

Soundness. Goals start at rank zero with compatible handover states. By induction, each promotion adds a safe state above a complete nonempty proving set of previously ranked winners: every UC successor, or every outcome of one control event. Queried buckets are immutable. Extraction retains all UC events, chooses one such control bucket at quiescence, and disables other controls. Every reached non-goal is safe, has a successor, and every successor has strictly smaller natural-number rank. Consequently no maximal run is an infinite run or a non-goal deadlock, and every run reaches a matching quiescent goal. The monitor rules advance exactly the memories active over each event; induction from entry establishes every declared safety interval. Handover retains the matching new memories.

Completeness. At failure both exploration queues are empty and promotion has reached a fixed point. Some entry is in $L = V \setminus W$, and L contains no goal. Coverage ensures that each safe state in L was expanded. If it is UC-enabled, at least one UC outcome lies in L , since otherwise it would be promoted. At quiescence every control candidate is exhausted; each enabled control bucket has an outcome in L for the same reason. Against any history-dependent policy the environment can select such an outcome after every history. Disabling all events deadlocks; otherwise the resulting run reaches an unsafe state or stays outside the goals forever. Hence no winning policy covers that entry. A winning policy therefore rules out failure and, by termination, forces success. Finally, UC priority loses no winning policy: removing optional controls at a UC-enabled state removes runs without introducing a deadlock, and the restricted policy remains admissible in the original game. $\square$

Checking the returned evidence. We write WIN for a completing policy and LOSS for proved nonexistence of one. WIN checking requires exactly $D_\sigma = \text{Reach}_\sigma(Q_0)$ as the rank domain, safety, enabled selected controls, nonempty all-outcome closure and strict decrease at non-goals. Goals require rank zero, quiescence and a compatible κ , defined exactly on $D_\sigma \cap Q_{\text{goal}}$; omitted policy entries are empty. LOSS checking requires an entry-containing, goal-free L with the UC/all-control counter-outcomes used in the completeness proof. These certify search relative to the successor and endpoint predicates [19]; Section 7 states their implementation trust boundary.

6.3 Exploring the Compiled Contract on Demand

With indexed predicates and constant-time bookkeeping, search is linear in discovered states, queried candidates and returned outcomes, excluding preprocessing and Java sorting/scans. Lazy may explore the whole reachable graph. For independent one-state updates, transfer-first Lazy visits $n + 1$ states versus Direct-Full's 2^n subsets; Technical Appendices K and O give the accounting and proof.

7 Implementation and Evaluation

7.1 Design and Implementation

RQ1 tests granularity; RQ2 checks correctness; RQ3 compares exploration and cost on 27 adapted contracts.

Each inherited input has three named requirement configurations: Base has no interval requirement; R1 protects selected actions during a requirement gap; R2 imposes selected ordering constraints. R1 blocks each designated action while its flag is true: any old-stop sets it, its designated new-start clears it. R2 requires selected boundaries before their transfers. In the GSM mobile-communication model, R1 blocks communication and decoding; R2 puts both decode-rule boundaries before transfer without ordering them. Event choices vary by model. PC2-Rolling, a production-cell calibration example, separately tests availability. The archived O-DUCS implementation (Legacy) also synthesizes subsequent control, a different objective.

Implementation. Our MTSA extension [11] checks certificates [29]; its composition checker, Link, checks the connected old, update and new controllers, including entries and targets. An independent oracle parses a restricted subset of Finite State Processes (FSP) [25]. Finite-JSON runs save losing membership; FSP benchmarks report only its statistics.

Use of generative AI. Generative AI assisted contract and example design, proof drafting, Java solver/frontend implementation, Python checkers, experiment drivers, execution, result analysis and manuscript editing. Validation used certificate and exhaustive checks for algorithms, source and automaton comparisons for initialization, saved-output checks for tables and primary sources for citations. Technical Appendix P details tool roles and validation; Threats discusses correlated faults.

Constructing the benchmark inputs. Nine inputs adapt eight DUCS sources. PC1 and PC2 are production-cell variants sharing one parent; they retain state-gated arrivals and make tool results UC. With work/output withheld, pending results and reset/arrival chains reach quiescence. Adaptations shorten Railcab transfers but retain both outcomes, omit two initial environment events in PowerPlant and alter safety/ordering. Equivalence is unestablished (Technical Appendix O).

MTSA synthesizes fixed endpoints from FSP and admits every reachable new-endpoint state. Before search, it builds monitors and global-tuple initializer tables; Update-interval (UPD) requirements use a constant post-entry state. Physical observers track nonlocal labels and survive commands/transfers. The finite-JSON Cell_n family, which extends Cell to n stations, supplies holder-dependent maps directly (S3). S4 reports all preparation and tuple counts.

Inputs have 1–5 components, 4–131 entries and 3–26 commands (Technical Appendix L). Only Railcab transfers branch; none declares precedence. R2 puts order in monitors; GSM/PC1 have one transfer, making its merge the identity.

Exploration variants. All methods share the same game and checks. On-the-fly variants apply UC priority and stop when all entries win; Direct-Full builds all buckets first. Full+UC instead prunes during construction. Lazy queries one control bucket at a time; Eager queries all. Update-first uses O-DUCS’s update priority [18]. When stops or transfers remain, Lazy prioritizes them before ordinary events and starts. Otherwise, it prioritizes starts only at endpoint alignment. It expands the most recently discovered state and retries unfinished states in arrival order. Technical Appendix K specifies all query-order ties.

Design and measures. For RQ3, one Windows/Xeon host uses a 64 GiB heap and a 1,200 s Java Virtual Machine (JVM) cap for 27×4 game cells plus 27 Legacy references. Medians require five completions; censored trials remain capped. Solver time includes checking; JVM adds frontend, startup, output and endpoints. Legacy uses its native timer; Mac pilots are excluded (Technical Appendices P, O.8; S4).

7.2 RQ1: Can Local Choices Change Feasibility?

Comparisons use Section 5’s merging rules; they do not estimate deployed prevalence.

Inherited inputs: no observed change. On nine inherited inputs, merging transfers, boundaries or both changes 0/54 Base/R1 decisions. Their total raw transfers omit Rolling’s startup phase; totality alone does not ensure safe merging. States decrease in 44 comparisons and stay equal in 10. R2 names individual events and is excluded. Source inspection supports Proposition 5.3 for 12/18 transfer-only pairs (Technical Appendix O.13); its one-way implication does not explain this null result.

Table 3. Granularity mechanisms, parameter sweeps and controls. Each pair compares fine and merged contracts of one model: Policy merges boundaries, other rows merge transfers. Counts are comparisons, not independent applications. The main total excludes Cell.

Family / control	Mechanism and parameters	Pairs	WIN / LOSS	LOSS / LOSS	WIN / WIN	Inc.
<i>Constructed families and parameter sweeps</i>						
Rolling	Readiness during startup. $n = 2 \dots 6, 1 \leq m < n$	15	15	0	0	0
Canary	Failure reports and recovery. $n = 2 \dots 6, 1 \leq m < n$	15	15	0	0	0
Policy	Audit overlap; role exclusion. 2 rule pairs	1	1	0	0	0
DB-Rolling	Role-gated replacement. $n = 3, m = 2, 3$	2	1	1	0	0
Rolling+Audit	Readiness and audit overlap. $n = 2, m = 1$; 1 audit pair	1	1	0	0	0
<i>Quiescence control</i>						
Threads (control)	Backpressure or persistent arrival. $n = 2 \dots 6$; queue 1,2	20	0	10	10	0
<i>Source-derived attempt</i>						
PC2-Rolling	Availability during calibration. 2 arms; floor 1	1	0	0	0	1
Main total		55	33	11	10	1
Cell _n (reference)	Empty-station transfer guards. $n = 2 \dots 10$	9	9	0	0	0

Outcomes read fine / merged; Inc. is unresolved. PC2’s merged solver remains TO, not LOSS. Rolling and Canary share a capacity obstruction: their sweeps supply 30 of the 33 WIN/LOSS pairs. n counts replicas/workers/stations. Canary bounds reported failures by $n - m$; otherwise m is an availability floor. Threads varies queue capacity 1,2 under both arrival regimes. Rolling+Audit’s boundary merge remains WIN and adds no pair here.

Constructed mechanisms. The families isolate local choices, retaining both-WIN and both-LOSS controls.

Table 3 records 33/55 fine-WIN/merged-LOSS comparisons across five constructed families. Of the 33 separations, 30 are Rolling/Canary parameter points, not independent mechanisms. Rolling compares $k = 1$ with $k = n$; its 70 threshold cells are separate. Threads tests both-WIN/both-LOSS quiescence controls. Technical Appendix O.11 gives all definitions and assumption controls.

Answer. Separate commands can preserve capacity, necessary intermediate stages and compatible requirement lifetimes. Merging can instead bypass a harmful mixed state (Section 5); finer is not uniformly better. Inherited decisions do not change. PC2’s source proof establishes merged impossibility, but its merged timeout leaves the solver comparison incomplete.

7.2.1 A Source-Derived Calibration Case. PC2-Rolling adds calibration while requiring another arm operational. Requests are controllable, completions UC, and transfers enter calibration while

preserving initializer observers. Erasing calibration recovers the original operational process, without whole-problem equivalence. Endpoints retain baseline requirements plus availability; the interval additionally treats transfers as calibration entry. The 25 boundaries and two transfers have no supplied order (Technical Appendix J).

The saved policy calibrates the arms successively, waiting for each completion. It covers all 126 entries; every outcome decreases rank from at most 37, with 1 arm outside calibration. Link checks composition. All thirteen source invariants satisfy the activation-scoped interpretation on reached tuples through their start-only suffixes. This covers neither all initializer tuples nor all 138 occurrences; GSM below exercises ordinary actions under an active new requirement.

For history-wide obligations, only 39/126 entries admit a safe full prefix; 87 fail (21 before entry, 66 during update). All admit a prior violating cycle restoring zero observers. Safe-history nonemptiness for 38,288 initializer tuples and runtime are unchecked (formulas, paths and partitions: Technical Appendix J).

Why the fixed merge loses. Every old entry admits both raw transfers. Before joint transfer, old plants lack calibration actions and all other steps preserve readiness. The joint transfer therefore puts both arms in calibration, clearing both interval readiness bits and reaching error. This proves impossibility, while the merged solver and fine Direct-Full both time out at 7200 s/200 GiB. Those timeouts remain unresolved solver results.

Restricted-transfer comparisons also retain both-LOSS and incomplete cases (Technical Appendix O; S1–S4).

7.3 RQ2: What Do the Checks Establish?

Answer. Table 4 maps the checks and their populations to the guarantee layers in Table 2.

Table 4. RQ2 validation populations, keyed to the scopes in Table 2.

Scope and check	Population	Established result and boundary
Game: independent games	46 fixtures, two solvers: 92 jobs	36 WIN, 48 LOSS, 8 invalid; all 84 eligible jobs pass certificate/exhaustive checks.
Invariant from start: New-requirement lookup tables	46 source positions, 138 occurrences; 900 non-error cells	Exact start-scoped languages under reference observations (Proposition 4.1); actual activation conformance unchecked.
From update entry: Interval initialization	135 occurrences: 46 R1, 89 R2	Exact entry-scoped initialization from source guards and stage fluents; execution is unchecked.
Game: application policies	25 Lazy WIN contracts; 125 runs	Certificates check the shared Java game. Runs save summaries; complete PC2-Rolling/Railcab exports are separate.

Independent reconstruction. For the 92-job suite, Python reconstructs explicit successors; front-end and endpoint/goal code are shared. All WIN jobs pass Link; invalid cases omit events, dead-lock or suppress UC. Six separate monitor-free fixtures independently reconstruct FSP endpoints, games and goals (S4.2).

From source invariants to initialized monitors. Initial truth and step checks establish monitor fidelity by induction; safe lookup reachability supplies Proposition 4.1’s premise (full checks in

Technical Appendix E.1). Actual decoding, post-start keys, observations and absence of fallback remain unchecked. The adapter preserves physical observers across commands, whereas reference action predicates can reset; conservativeness is unproved. These checks validate neither arbitrary FSP compilation nor source semantics for all WIN policies.

Which history is meant. All 138 new-requirement (NEW) occurrences also satisfy residual inclusion for safe prior histories; actual scope nonemptiness remains unverified except one PC2 tuple. UPD monitors instead begin their reference at update entry (Technical Appendix D).

GSM after activation. For GSM’s change from $\square\text{-decode.3}$ to $\square\text{-decode.2}$, a source-derived policy covers every entry. It starts the new requirement with observer/monitor zero, permits *decode.3* and output before transfer, and rejects *decode.2*. It satisfies R1’s gap restriction and reaches a matching endpoint within twelve events, proving ACT over ordinary behavior. Technical Appendix E gives the path, rank proof and separate 45-pair entry-initialization check.

What application WIN and LOSS mean. Railcab’s Base policy covers 22 entries with 1–4 ordinary gap events; R1’s has none and retains seven interval guards. Both pass certificate/Link. WIN alone supplies no gap duties; explicit interval requirements let the same synthesis check a gap-protection policy (Technical Appendix O.9).

One Base milestone transfer has two outcomes (S4.4.3): from rank 13, one reaches rank 11 and needs 11 more events to new-controller state 3; the other reaches rank 12 and needs 12 more events to state 1. Both must complete.

Exhaustive closure shows Industry/R1 loses at 42/131 entries under every order: validating either of two components violates new requirements; another component may reject forever, or deadlock quiescently if safe validation is withheld. Native DUCS wins its different subsequent-control problem (Technical Appendix O; S4).

Across five repeats, Lazy’s WIN certificates consistently have 8–1,071 states and maximum ranks 4–34; linked controllers have 15–1,260 states (S4). Ranks bound events, not downtime or optimal length (Technical Appendix O.8); they certify model-level progress.

7.4 RQ3: Does Lazy Reduce Exploration?

Answer. With the same UC pruning, Lazy decides 26 contracts to Eager’s 17. Direct-Full additionally changes pruning and decides 14; Lazy reduces states and queries in all 14 completed pairs. All paired decisions agree (Table 5). Update-first is faster on 8/24 paired solver medians but adds two timeouts. Lazy returns LOSS for Industry/R1 and times out on PC2/R2 at 1,200 s/64 GiB (fixed), 3,600 s/64 GiB (sensitivity) and 7,200 s/200 GiB (S4, earlier series); the limiting resource is unknown.

Legacy returns 9 WIN, 13 out-of-memory (OOM) failures and 5 unmeasured snapshot-cap cases. Because it also synthesizes subsequent control, these are unmatched references (Technical Appendix O).

Figure 2 shows all 14 completed pairs. State reductions are largest in Base and smaller in R2; these different completed populations do not isolate interval effects or attribute savings to requirement gaps (Technical Appendix M). In R2, Update-first has fewer states and lower solver medians in six of eight pairs (Technical Appendix O).

Lazy’s adapter medians reach 14.221 s (PC2/R1; S4). Capped Direct-Full JVM totals are lower bounds, not solver medians or ratios. Auxiliary Full+UC timings mix builds, hosts and repeats, preventing pruning isolation; unpruned OTF is unimplemented (Technical Appendix O.10).

Scaling limits. All methods finish the independent family for $n = 2, \dots, 20$. At $n = 20$, Lazy and Update-first discover 21 states each, Eager 211, and Direct-Full 1,048,576 (S4).

Table 5. All 27 contracts per method. TO denotes one capped trial, not a timing median. Ratios summarize heterogeneous completed pairs; values above one favor Lazy.

Method	WIN/LOSS/TO	Faster / pairs	Solver ratio	Total JVM ratio
Lazy	25/1/1	—	—	—
Eager	17/0/10	0 / 17	2.04 [1.18–4,575.05]	1.60 [1.00–144.85]
Update-first	23/1/3	8 / 24	1.07 [0.19–39.76]	1.00 [0.32–6.10]
Direct-Full	14/0/13 [†]	0 / 14	9.14 [1.58–2,123.81]	2.27 [1.00–130.66]

TO: timeout; no OOM/unmeasured cases. [†] Workflow/R2 retains its interruption and later same-setting timeout (S4). Comparator/Lazy ratios are median [min–max] across completed pairs of five-run medians. Faster / pairs counts lower comparator solver medians among all completed pairs. JVM includes frontend, startup, endpoints and output; solver includes checking. Unresolved costs have no ratio.

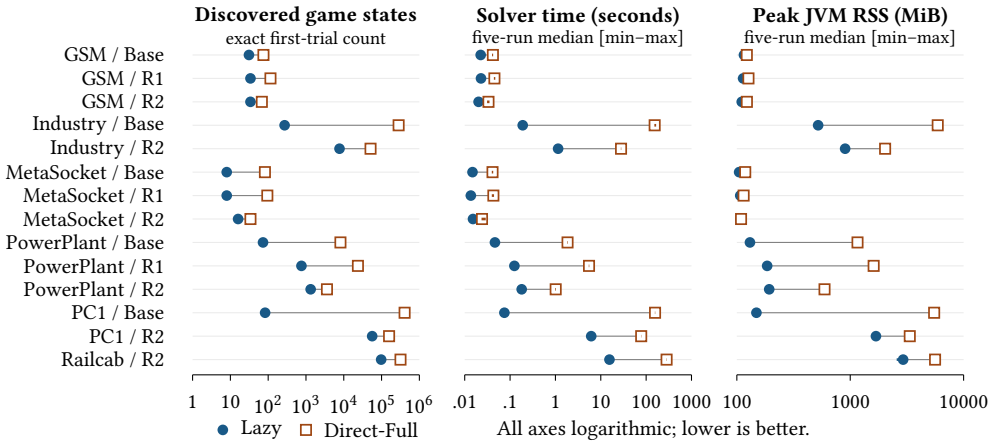


Fig. 2. All 14 completed WIN/WIN pairs. Solver time includes checking but excludes frontend, endpoint preparation and JVM startup. Resident set size (RSS) is sampled over the whole JVM run; it is neither solver-only nor heap usage. The other 13 Direct-Full contracts time out (Figure A4) and are omitted. This compares whole procedures.

Travel reaches timeouts and out-of-memory failures under all methods, including expanded-budget timeouts. These separate trials never enter medians; Technical Appendix O.10 retains all populations and adverse outcomes.

8 Related Work

Generating update environments. Update-environment generation from supplied global state-preservation transitions precedes FG-DUCS [44, Sec. V]. DUCS adds reconfigure edges through possibly partial R , without a UC-quiescence guard [34, Sec. 4.2, Defs. 3.7, 5.1]. FG-DUCS requires global UC quiescence at commands and handover.

On-demand synthesis. Directed controller synthesis (DCS) already combines lazy products and controls, UC priority and backward propagation [9, Sec. IV]. Its 2016 formulation requires safety and eventual goal occurrence on every trace; the later nonblocking formulation (OTF-DCS, 2023) requires a marked extension of each finite controlled trace [10, Def. 4]. FG-DUCS specializes these

patterns to its contract, with full outcome buckets and all-entry certificates. Other approaches structure specifications compositionally or modularly [31, 43].

Supervisor switching. Related guarantees include safe, deadlock-free supervisor switching [30, Sec. 3.3], workcell switching faster than UC events [20, Sec. III], startup bounds under weak invariance [35] and mode-specific invariants under UC switches [36].

Requirements across adaptation. Obligations across change occur in temporal adaptation, scenario updatability and Live Synthesis [13, 14, 27, 45]; FG-DUCS uses scoped requirements and residual reasoning [12, 37]. Thuijsman and Reniers [40, Secs. 4–7] support individual/simultaneous re-configuration, feature-dependent requirements and state retention/reset. Their guarantee requires a marked continuation from every reachable state; FG-DUCS requires all maximal update runs to finish at a supplied controller.

Safe adaptation paths. Safe adaptation graphs support minimum-cost paths [46, Secs. 3–4]. Re-configuration sequences preserve lifecycle and architectural constraints, reaching the target without failures or a consistent architecture after recovery [5, Secs. II–VI]. FG-DUCS instead checks finite handover to a fixed controller under declared requirement lifetimes; it does not optimize cost or repair failures.

Operational update mechanisms. Local, live, workflow and network updating address migration and transition safety [15, 16, 26, 28, 38]. Dynamic dependencies and sufficient freeness support transaction version consistency [24]; FG-DUCS’s finite handover does not imply that guarantee.

9 Threats to Validity

Perpetual UC arrivals or polling block local commands; Threads contrasts finite UC phases with perpetual activity. Fairness-dependent waiting is outside the contract; inherited-adaptation effects on feasibility are not isolated. Decoding, history completeness and A1–A4 remain unvalidated; repair is excluded. Constructed cases assume finite startup, reporting, recovery and election; selection may favor separations, and deployed prevalence is unestablished.

Shared authorship, AI and code risk correlated faults; independent parsing covers restricted FSP. Direct-Full is in-house and Legacy solves a different objective. External solvers, pruning isolation and randomized orders are untested. Ratios exclude unresolved costs; extended trials lack medians. Sampled RSS can miss peaks and identifies neither heap use nor bottlenecks; S4 retains interruptions.

Null/LOSS results, faster Update-first, lower Eager/Update-first memory and Travel/PC failures bound gains; PC2’s merged timeout remains unresolved. One-machine fixtures/repeats are not independent applications. Inherited inputs lack precedence and only Railcab transfers branch. Exponential separation assumes the default order.

10 Conclusion

FG-DUCS compiles local transfers, requirement lifetimes and initializers into safe finite updates preserving the selected endpoint’s continuation. Merging can help or prevent completion; a local UC condition guarantees safe serialization of transfer-only merges. With the same UC pruning, Lazy decides 26/27 contracts to Eager’s 17; Update-first is sometimes faster. Guarantees require declared lifetimes, UC quiescence and faithful execution. This supports maintenance planning with explicit duties and checkable policies.

Data Availability

We created and publicly released the anonymous FG-DUCS replication package for this paper [4]. The retained evidence snapshot is available at <https://github.com/research-artifact-archive/research-artifact/tree/fgducs-invariants-20261002>. The paper-matched release cited in [4], fgducs-c103-20261003, contains this revised main paper and paper/supplementary_material.pdf: Part I is Technical Appendix A–P; Part II is Supplement S1–S5. Its reading guide, bookmarks and continuous page numbers locate the proofs, full models and experimental records cited throughout the main text. The repository README links directly to both parts and maps paper sections to evidence. The package also provides source code, all 27 application contracts, scripts and saved results; saved-evidence checks require no new performance experiments. Dependencies come from official upstream sources.

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